

Transformers.

Transfer learning $\rightarrow$ multimodal task $\rightarrow$ NLP + Image

* In Attention mechanism the drawback or the disadvantage is, we can't give the parallel seq all the words in a sentence $\rightarrow$ Not Scalable.

Dataset $\rightarrow$ huge $\rightarrow$ Scalable with respect to training.

Transformers $\neq$ LSTM RNN

$\rightarrow$ contains self-attention module
 $\rightarrow$ All the words will parallelly sent to encoder.

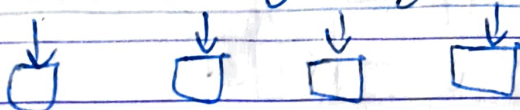
$\Downarrow$
Positional Encoding.

Transformer $\rightarrow$ $\uparrow\uparrow\uparrow$ Dataset $\uparrow\uparrow\uparrow$ $\rightarrow$ Amazing
SOTA $\leftarrow$ NLP

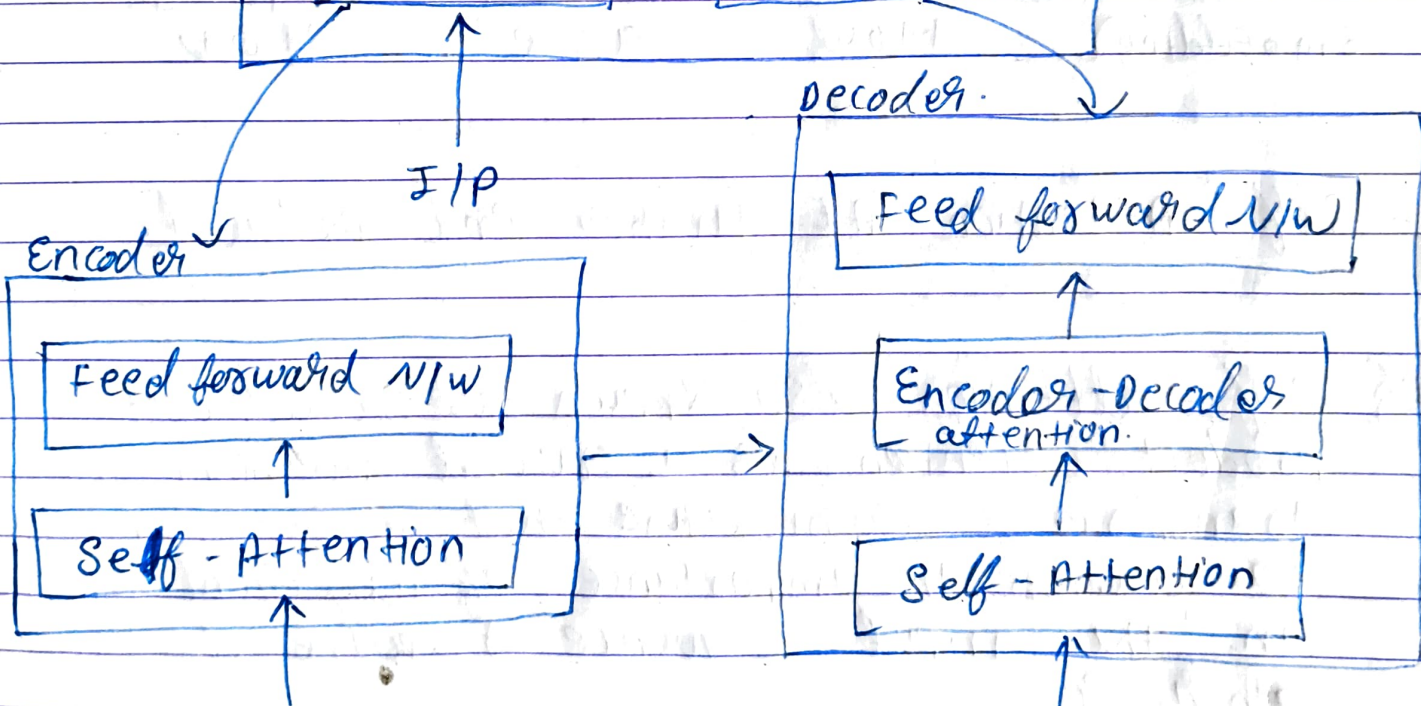
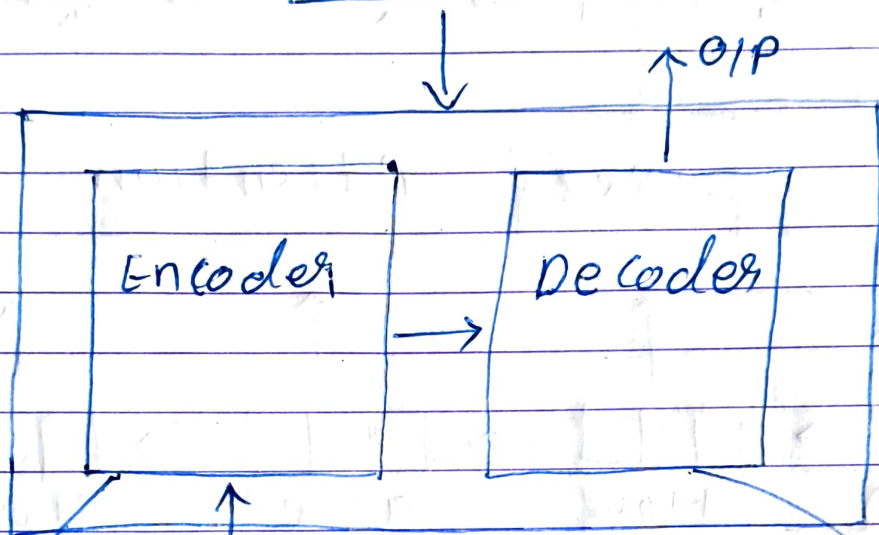
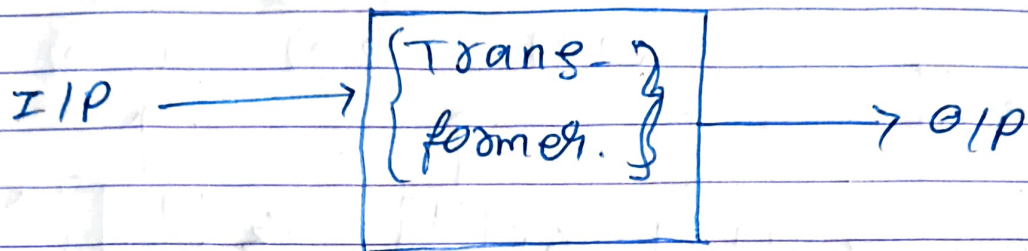
Contextual Embedding $\Rightarrow$ related.

my name is vidya, I play cricket.

$\downarrow$
[Embedding Layer]

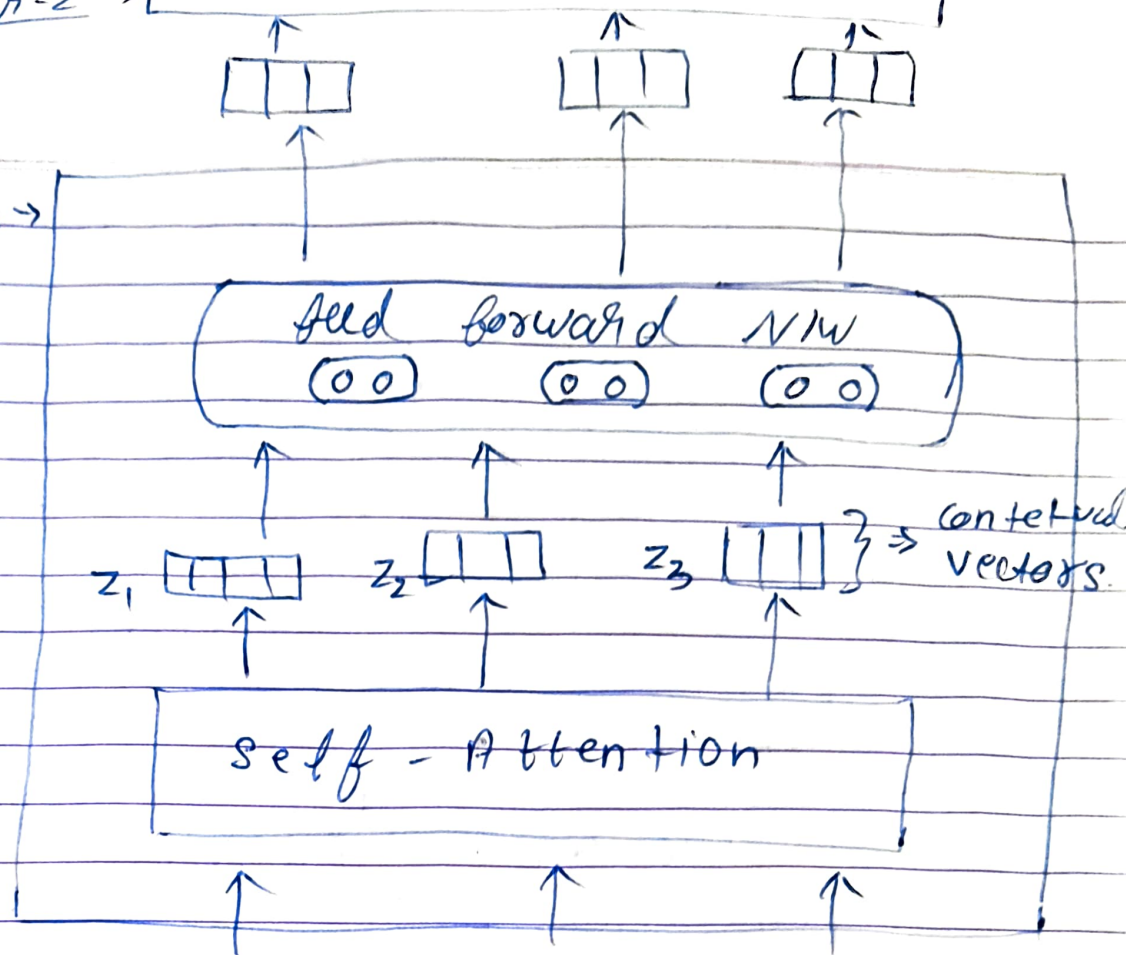


Basic Architecture of Transformer



Encoder-2 $\rightarrow$ L

Encoder-1 $\rightarrow$



Vectors $\rightarrow$ $\begin{bmatrix} \square & \square & \square \end{bmatrix} \rightarrow \begin{bmatrix} \square & \square & \square \end{bmatrix} \rightarrow \begin{bmatrix} \square & \square & \square \end{bmatrix}$
(Embedding) How are you

$\rightarrow$ Self Attention At Higher And Detailed.

$\Rightarrow$ { Self-attention, also known as scaled dot-product attention, is a crucial mechanism in the transformer architecture that allows the model to weigh the importance of different tokens in the input sequence relative to each other. }

~~Idea~~ Input $\Rightarrow$ Queries (Q), Keys (K),
values (V).

① Query vectors (Q):

It represents the token for which we are calculating the attention. They help determine the importance of other tokens in the context of the current token.

Importance $\Rightarrow$

- ① focus determination.
- ② contextual understanding.

② Key vectors (K):

This represents all the tokens in the sequence and are used to compare with the query to calculate attention scores.

Importance $\Rightarrow$

- ① Relevance measurement
- ② Information Retrieval.

③ value vectors (V):

It is held the actual information that will be aggregated to form the output of the ~~calculation~~ attention mechanism.

Importance $\Rightarrow$

- ① Information Aggregation.
- ② Context preservation.

Input Sequence = ["The", "cat", "sat"]

Embedding size = 4 , $Q, K, V \Rightarrow 4$

① Token Embedding $\Rightarrow$

$$E_{\text{The}} \Rightarrow [1 \ 0 \ 1 \ 0]$$

$$E_{\text{cat}} \Rightarrow [0 \ 1 \ 0 \ 1]$$

$$E_{\text{sat}} \Rightarrow [1 \ 1 \ 1 \ 1]$$

② Linear Transformation $\Rightarrow$

we create Q, K, V by multiplying the embeddings by learned weights matrices w_Q, w_K, w_V .

$$w_Q = w_K = w_V \Rightarrow I \Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$Q_{\text{The}} \Rightarrow [1 \ 0 \ 1 \ 0] \cdot w_Q \Rightarrow [1 \ 0 \ 1 \ 0]$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$K_{\text{The}} = [1 \ 0 \ 1 \ 0] \cdot w_K = [1 \ 0 \ 1 \ 0]$$

$$V_{\text{The}} = [1 \ 0 \ 1 \ 0] \cdot w_V = [1 \ 0 \ 1 \ 0]$$

$$\textcircled{i} Q_{\text{The}} = K_{\text{The}} = V_{\text{The}} \Rightarrow [1, 0, 1, 0]$$

$$\textcircled{ii} Q_{\text{cat}} = K_{\text{cat}} = V_{\text{cat}} \Rightarrow [0, 1, 0, 1]$$

$$\textcircled{iii} Q_{\text{sat}} = K_{\text{sat}} = V_{\text{sat}} \Rightarrow [1, 1, 1, 1]$$

③ Computing Attention Scores: $\Rightarrow$

$$\textcircled{i} \underset{\substack{\downarrow \\ \text{token}}}{\text{The}} \Rightarrow \text{score}(Q_{\text{The}}, K_{\text{The}}) = [1 \ 0 \ 1 \ 0] \cdot [1 \ 0 \ 1 \ 0]^T$$

$$\Rightarrow [1 \ 0 \ 1 \ 0] \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = 2$$

$$\text{score}(Q_{\text{The}}, K_{\text{cat}}) = [1 \ 0 \ 1 \ 0] \cdot [0 \ 1 \ 0 \ 1]^T \Rightarrow 0$$

$$\text{score}(Q_{\text{The}}, K_{\text{sat}}) = [1 \ 0 \ 1 \ 0] \cdot [1 \ 1 \ 1 \ 1]^T = 2$$

② cat $\rightarrow$ Token

$$\text{score}(Q_{\text{cat}}, K_{\text{The}}) = [0 \ 1 \ 0 \ 1] \cdot [1 \ 0 \ 1 \ 0]^T = 0$$

$$\text{score}(Q_{\text{cat}}, K_{\text{cat}}) = [0 \ 1 \ 0 \ 1] \cdot [0 \ 1 \ 0 \ 1]^T = 2$$

$$\text{score}(Q_{\text{cat}}, K_{\text{sat}}) = [0 \ 1 \ 0 \ 1] \cdot [1 \ 1 \ 1 \ 1]^T = 2$$

iii) for the token sat.

$$\text{Score}(Q_{\text{sat}}, K_{\text{re}}) = [1 \ 1 \ 1 \ 1] \cdot [1 \ 0 \ 1 \ 0]^T = 2$$

$$\text{Score}(Q_{\text{sat}}, K_{\text{cat}}) = [1 \ 1 \ 1 \ 1] \cdot [0 \ 1 \ 0 \ 1]^T = 2$$

$$\text{Score}(Q_{\text{sat}}, K_{\text{sat}}) = [1 \ 1 \ 1 \ 1] \cdot [1 \ 1 \ 1 \ 1] = 4$$

iv) Scaling $\Rightarrow$

we take up the scores and scale down by dividing the scores by the $\sqrt{(\text{dim})_K}$

$$\Rightarrow \sqrt{(\text{dim})_K} = \sqrt{(K)_{\text{dim}}} \Rightarrow \sqrt{d_K} = \sqrt{4} = 2$$

scaling in the attention mechanism is crucial to prevent the dot product from growing too large $\Rightarrow$

It ensures stable gradients during training

d_K is large $\rightarrow$ ① Gradient Exploding.

two problems we are facing $\rightarrow$ ② softmax saturation.

Let $\Rightarrow$

$$Q = [2 \ 3 \ 4 \ 1] \quad , \quad K_1 = [1 \ 0 \ 1 \ 0]$$

$$K_2 = [0 \ 1 \ 0 \ 1]$$

without scaling.

$$Q \cdot K_1^T = 2 \times 1 + 3 \times 0 + 4 \times 1 + 1 \times 0 \Rightarrow 2 + 4 = \underline{6}$$

$$Q \cdot K_2^T = 2 \times 0 + 3 \times 1 + 4 \times 0 + 1 \times 1 \Rightarrow 0 + 3 + 0 + 1 = 4$$

* Scores $[6, 4] \Rightarrow$ without scaling.

applying softmax. ($[6, 4]$)

$$\Rightarrow \left[\frac{e^6}{e^6 + e^4}, \frac{e^4}{e^6 + e^4} \right]$$

$$\Rightarrow \left[\frac{e^6}{e^6(1+e^{-2})}, \frac{e^4}{e^4(1+e^2)} \right]$$

$$\Rightarrow \left[\frac{1}{(1+e^{-2})}, \frac{1}{e^2+1} \right] \Rightarrow \approx [0.88, 0.12]$$

$\Rightarrow$ most of the attention weight is assigned to the first Key vector, very little to the second vector,

$\Rightarrow$ with scaling.

① compute scaled dot product.

$$[6, 4] \Rightarrow \text{Scaled} \Rightarrow \left[\frac{6}{\sqrt{d_K}}, \frac{4}{\sqrt{d_K}} \right]$$

d_K = dimension of (K)

$$\sqrt{d_K} = \sqrt{4} = 2$$

$$[6, 4] \Rightarrow \left[\frac{6}{2}, \frac{4}{2} \right] = [3, 2]$$

$$\text{Softmax} \Rightarrow \left[\frac{e^3}{e^3 + e^2}, \frac{e^2}{e^3 + e^2} \right]$$

$$\Rightarrow \left[\frac{\cancel{e^3}}{e^3(1+e^{-1})}, \frac{\cancel{e^2}}{e^2(e+1)} \right]$$

$$\Rightarrow \left[\frac{1}{1+e^{-1}}, \frac{1}{e+1} \right] \approx [0.73, 0.27]$$

$\Downarrow$
attention weights.

{ Here, the attention weights are more balanced compared to the unscaled case.

Imp Scaling ensures that the dot products are kept within a range that allows the softmax function to operate effectively, providing a more balanced distribution of attention weight and improving the overall learning process the model

~~Score~~ Score for "The" $\Rightarrow [1, 0, 1]$

~~Score~~ Score for "cat" $\Rightarrow [0, 2, 2]$

~~Score~~ Score for "sat" $\Rightarrow [2, 2, 4]$

⑤ Apply softmax.

Attention weight "The" = $\text{Softmax}([1, 0, 1])$

$\Rightarrow [0.4223, 0.1554, 0.4223]$

Attention weight "cat" = $\text{softmax}([0, 2, 2])$

$\Rightarrow [0.1554, 0.4223, 0.4223]$

Attention weight "sat" $\Rightarrow \text{Softmax}([2, 2, 4])$

$\Rightarrow [0.2119, 0.2119, 0.5762]$

⑥ weight sum of values $\Rightarrow$

we multiply the attention weight by corresponding value vectors. ~~for the token~~

For the Token "The" =

$$= (0.4223 * V_{\text{The}}) + (0.1554 * V_{\text{cat}})$$

$$\Rightarrow \{0.4223 \cdot [1 \ 0 \ 1 \ 0] + 0.1554 \cdot [0 \ 1 \ 0 \ 1] + 0.4223 \cdot [1 \ 1 \ 1 \ 1]\}$$

$$\Rightarrow [0.4223, 0, 0.4223, 0] + [0, 0.1554, 0, 0.1554] + [0.4223, 0.4223, 0.4223, 0.4223]$$

$$\Rightarrow [1.2669, 0.999, 1.2669, 0.999]$$

The $\rightarrow [1 \ 0 \ 1 \ 0] \rightarrow$ embedding vector

$\downarrow$
after the self attention model or the layer the value becomes.

$\downarrow$
[1.2669, 0.9999, 1.2669, 0.9999]
 $\rightarrow$ Known as the contextual vector

$\rightarrow$ process for finding the contextual vector.

- (i) take the embedded vector
- (ii) Initialize Q, K, V, w_Q, w_K, w_V
- (iii) finding the Attention score
- (iv) scaled the score
- (v) apply the softmax and find the values.
- (vi) calculate the weighted sum of value ($\text{softmax} \times V_i$)

Sequence to Sequence mechanism.

